^{36}Ar(n,d) 1977Pa29

$J^{\pi}=0^{+}$ for ^{36}Ar ground state.

1977Pa29: 14.1-MeV neutrons were produced via the ^{3}H(d,n)^{4}He reaction at an average rate of 10^{9} neutrons per second into 4π sr from the Brown University neutron generator and impinged on a 2.85-mg/cm^{2} ^{36}Ar target in the form of a gas cell filled with 99.8% enriched ^{36}Ar. Deuterons were detected using a time-of-flight counter telescope consisting of two proportional counters and a NaI(Tl) crystal. Measured $\sigma(E_d,\theta)$. Deduced levels, L-transfers, spectroscopic factors from local zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

^{35}Cl Levels

E(level)†	L‡	Comments
0	2	S: 3.83 *48* and 3.90 *47* using two sets of optical parameters.
1219#	0#	S: 1.94 *23* and 3.30 *40* using two sets of optical parameters. Other spectroscopic factors calculated using two sets of optical parameters did not show large differences.
1763#	(2)#	

† From 1977Pa29.

‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1977Pa29.

\# 1977Pa29 state that the first and second excited states are recorded together and not resolved, but their angular distributions could be separated due to different L-transfers, however statistical errors prevent an accurate determination of the smaller L=2 component from the second excited state.